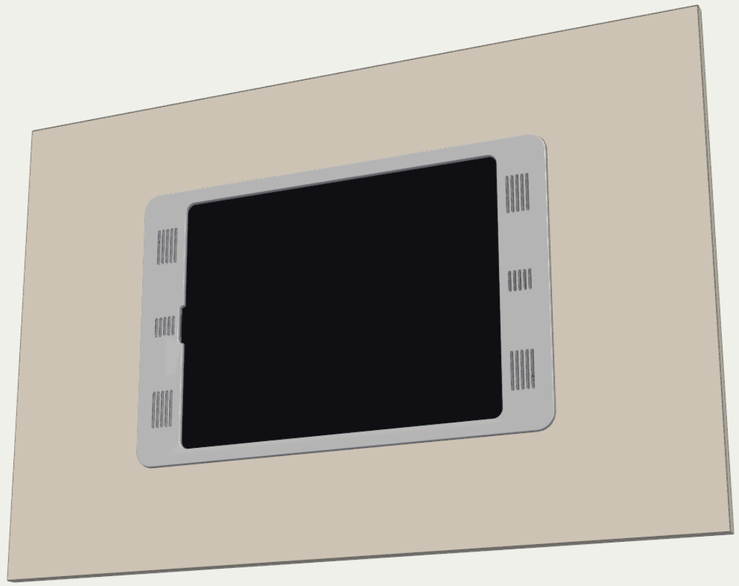
USER MANUAL · V5.1 FLUSH

FLOAT MOUNT

In-wall PoE mount for iPad Pro 12.9″ (3rd gen)

A 3D-printed box set into the plasterboard, finished with a 3 mm machined 6061 trim. The glass sits almost flush with the wall.



Designed for Brendan · October 2026 · Read the whole manual before you cut the wall

CONTENTS

WHAT'S IN THIS MANUAL

General data and design features	3
Exploded view and cross-sections	5
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Print settings, parts list and files	11
Parts orientation	14
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Bench assembly (10 steps)	19
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Removing the iPad	38
iPad setup and final checklist	39

Colour code used in every step: **orange** = the part you are working on now, **blue** = screws, inserts, magnets and other hardware, grey = parts already fitted.

OVERVIEW

GENERAL DATA

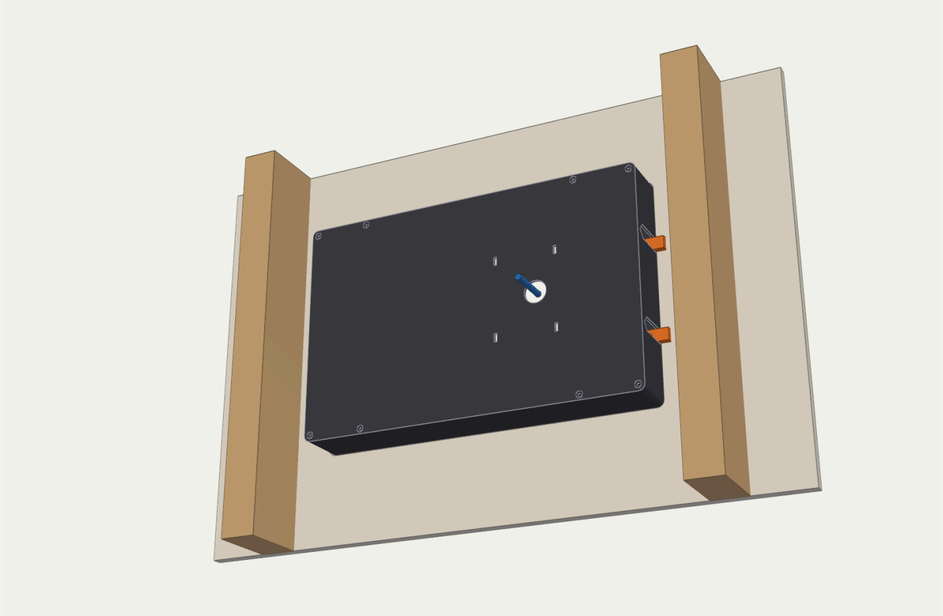
Item	Value
Trim size	355.1 × 233.6 × 3 mm 6061, ~188 g
Trim width	ends 37.2 mm · top/bottom 9.3 mm
Off the wall	trim 4.5 mm · glass 1.2 mm
Wall cut-out	343.1 × 221.6 mm, R7 corners
Depth in the wall	57.5 mm from the wall face
Wall	10 mm plasterboard, 70 mm studs at 450 c/c
Fits	iPad Pro 12.9″ 3rd gen (A1876/A1895/A2014/A1983)
Power + data	PoE+ (802.3at) over one Cat6
Fixing	4 dog-ear arms clamp the plasterboard
Printer	256 × 256 mm bed (P1S / X1C class)



6061 trim, front view. Orange dashed = plasterboard cut-out 343.1 x 221.6 mm (hidden). Proud of the wall 4.5 mm.

OVERVIEW

DESIGN FEATURES



FLUSH GLASS

Only the 3 mm trim stands proud. The glass is 1.2 mm off the wall.

NO STUD WORK

Fits between studs in existing plasterboard. No noggings, no patching.

MAGNETIC TRIM

8 magnet pairs and 2 pins. No screws on the face at all.

DOG-EAR CLAMP

4 arms swing out behind the board. The screws also lock the deck.

PoE IN THE BOX

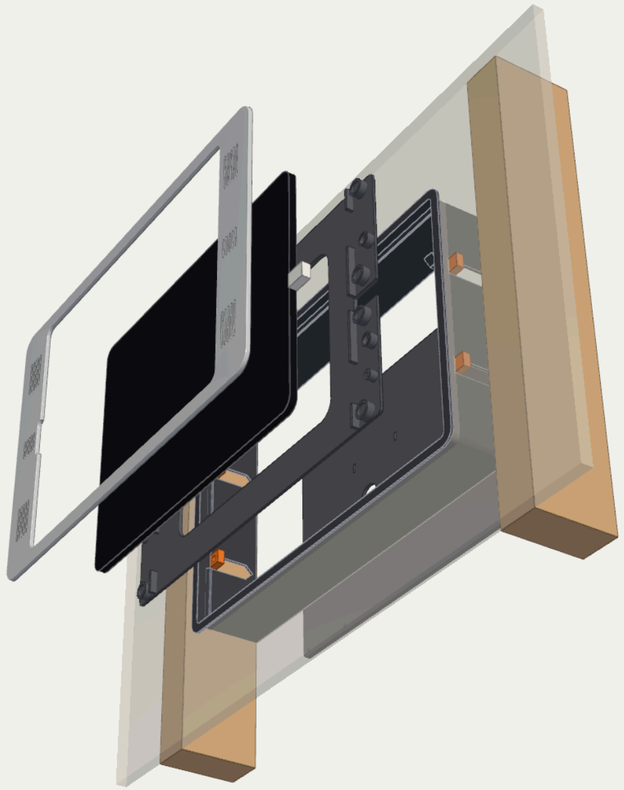
The adapter rides on the back plate. One Cat6 in, nothing else.

NO SUPPORTS

Every printed part prints flat, with 45° ledges instead of overhangs.

OVERVIEW

EXPLODED VIEW



1 **TRIM** · 6061, magnets + pins

2 **iPAD PRO** + straight USB-C

3 **DECK** L + R · carries the iPad

4 **SLEEVE** L + R · box and flange

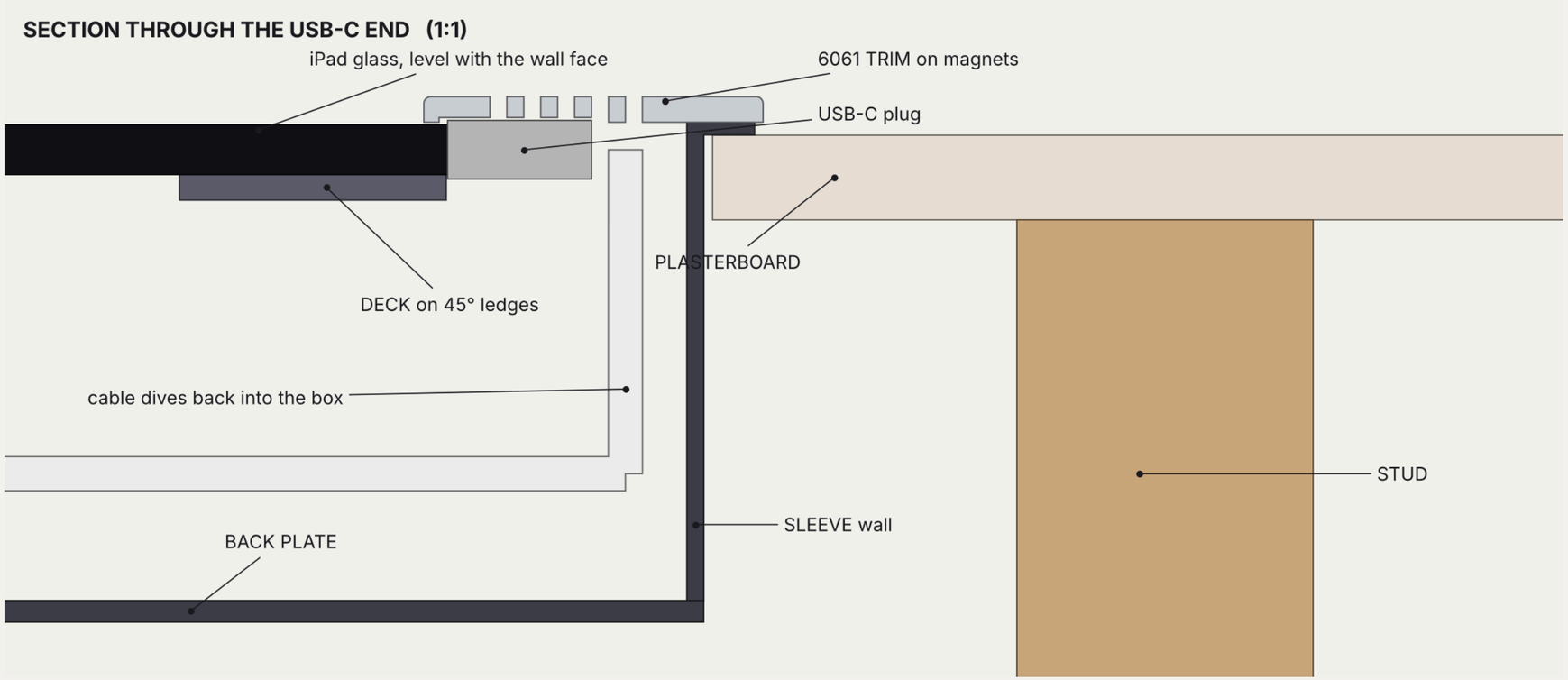
5 **DOG-EAR ARMS** × 4

6 **PoE ADAPTER**

7 **BACK PLATE** L + R

OVERVIEW

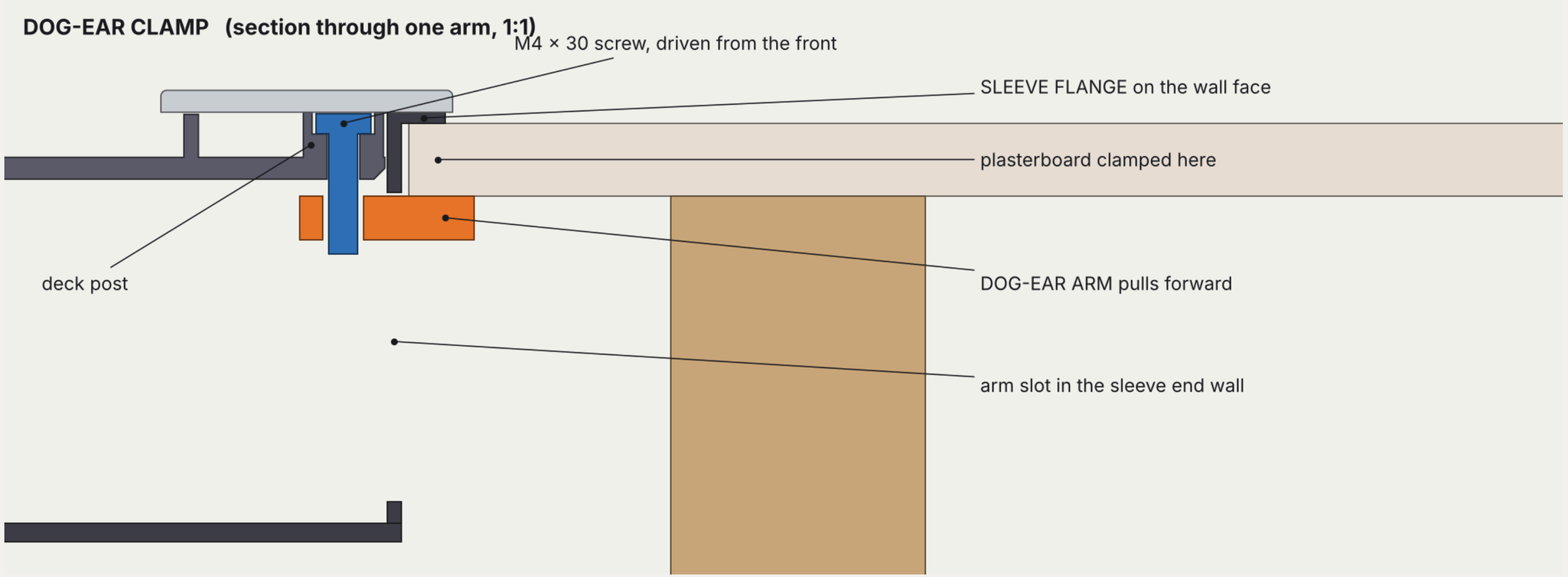
HOW IT SITS IN THE WALL



A true 1:1 cut through the USB-C end. The trim lip holds the glass through 0.3 mm PORON tape. The plug and its bend live inside the box end, under the wide part of the trim, and the cable drops to the PoE adapter on the back plate.

OVERVIEW

HOW IT CLAMPS



Each **M4 × 30** runs from the DECK POST down the inside of the sleeve wall into an insert in a DOG-EAR ARM, which reaches out through a slot in the wall. Tighten it and the arm pulls up behind the plasterboard, squeezing the board between the arm and the 6 mm FLANGE. The screw head also holds the deck down on its ledges.

BEFORE YOU START

RECOMMENDED EQUIPMENT

Item	Notes
PoE+ switch port	802.3at (PoE+). One port per panel.
PoE+ to USB-C adapter	PoE Texas / Heckler T515, 25 W. Lives inside the box.
USB-C cable	Standard straight plug, short. Box end sized for a 17 × 12.5 × 7 mm overmold.
Cat6 cable	From the switch into the stud bay, with about 300 mm spare at the hole
Filament	PETG or ASA. Not PLA: the iPad warms a closed box.
Aluminium	6061 plate, 3 mm (or 4 mm faced down), at least 360 × 240 mm
Magnets	N52 discs, 10 × 1.5 mm, 16 off
PORON tape	0.3 mm thick, about 1 m, for the trim lip that touches the glass

BEFORE YOU START

REQUIRED HARDWARE

Item	Qty	Where it goes
N52 magnet 10 × 1.5 mm	16	8 in the trim pockets · 8 in the deck posts
Dowel pin Ø3 × 6 mm	2	Pressed into the trim, locate in the deck
M4 × 30 pan-head screw	4	Deck posts down to the dog-ear arms
M4 heat-set insert	4	One in each dog-ear arm
M3 × 8 countersunk screw	8	Back plate to the sleeve
M3 heat-set insert	8	Bosses on the sleeve’s back edge
2-part epoxy	1	Sleeve seam, deck seam and magnets
Zip ties, small	2	PoE adapter to the back plate
PORON tape 0.3 mm	1 m	Underside of the trim lip

BEFORE YOU START

TOOLS

BENCH

- Soldering iron + M3/M4 heat-set tips
- Flat surface (glass or MDF)
- Spring clamps, masking tape
- Screwdriver for the M4 pan heads
- 2 mm hex key (M3 countersunk)
- Isopropyl alcohol + cloth

MACHINING

- CNC router (StrongHold)
- 6 mm and 2 mm end mills
- 3 mm drill
- Vacuum table or double-sided tape on a spoilboard
- Deburring tool

WALL

- Stud finder + spirit level
- A3 cut-out template
- 14 mm drill bit
- Plasterboard saw
- Torch + inspection mirror
- Draw wire for the Cat6

PRINTING

PRINT SETTINGS

Setting	Value
Material	PETG or ASA
Layer height	0.2 mm (0.4 mm nozzle)
Walls	4 (6 on the dog-ear arms)
Infill: sleeve	15% gyroid
Infill: deck, back plate	20% gyroid
Infill: dog-ear arms	100%
Supports	None needed
Bed	Textured PEI

No supports. Overhang audit in print orientation: sleeve 0 mm², deck 3 mm², back plate 21 mm², arms 0 mm².

The 21 mm² is harmless. It is the countersinks on the back plate’s first layer.

The arms take the clamp load. Print them solid.

PRINTING

PARTS LIST

Part	Qty	Material	Size
TRIM	1	6061 aluminium	355.1 × 233.6 × 3 mm
SLEEVE L, R	2	PETG / ASA	176.6 × 231.6 × 56.5 mm each
DECK L, R	2	PETG / ASA	168.3 × 215 × 9.2 mm each
BACK PLATE L	1	PETG / ASA	230.6 × 219.6 × 2.5 mm
BACK PLATE R	1	PETG / ASA	110.6 × 219.6 × 2.5 mm
DOG-EAR ARM	4	PETG / ASA	24 × 12 × 6 mm

The back plate is split off-centre on purpose: its joint sits 60 mm from the sleeve seam, so the plate bridges and stiffens that seam.

PRINTING

FILES

print/ (STL + STEP)

- sleeve_L · sleeve_R
- deck_L · deck_R
- back_plate_L · back_plate_R
- dog_ear_arm_x4 (all four on one plate)

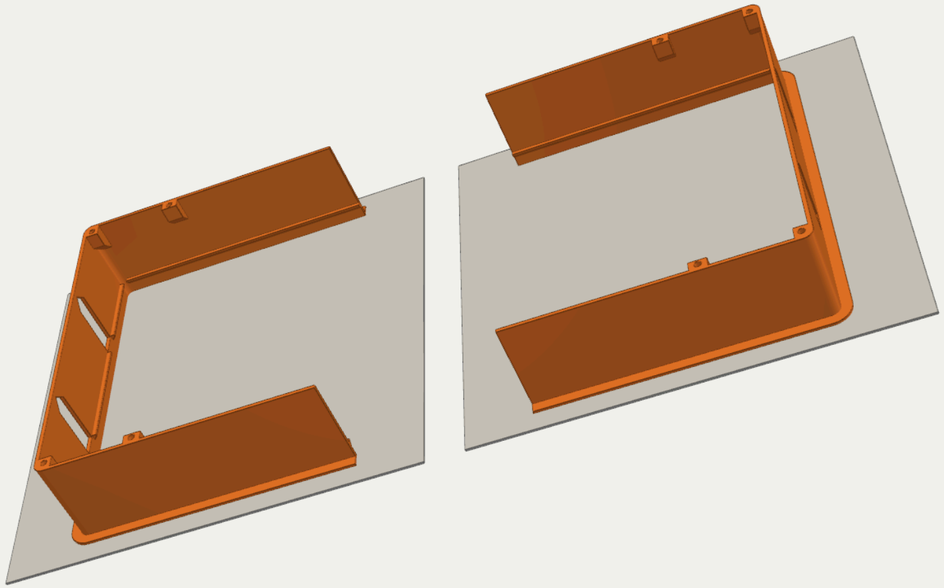
machine/, CAD and wall

- machine/trim_6061.step for CAM
- assembly_flush_v5_1.step (whole stack)
- CUTOUT_TEMPLATE_A3.pdf (print at 100%)
- flush51.py: the parametric source
- BUILD_NOTES.md

Every size in this manual comes from flush51.py. Change a parameter there (plug size, plasterboard thickness, clearance) and re-run it to regenerate the model files. If the cut-out size changes, make a new template too.

PARTS ORIENTATION · 1 OF 3

SLEEVE HALVES



Face down, as exported. One half per plate.

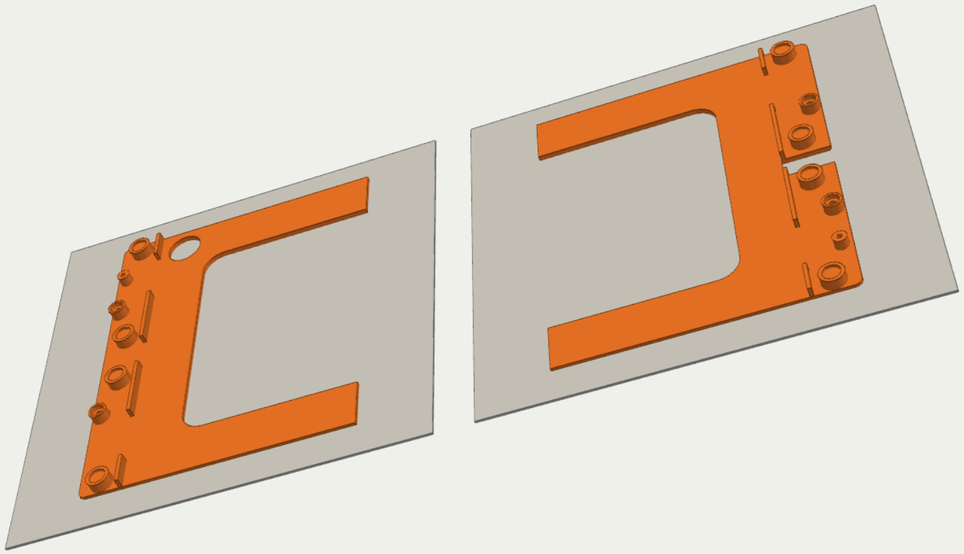
The flange face goes on the bed, so the surface that sits against the wall comes out dead flat.

The ledges inside are 45°, and the arm slots have 45° roofs, so nothing needs support.

> 176.6 × 231.6 × 56.5 mm. Fits a 256 mm bed with room for a brim.

PARTS ORIENTATION · 2 OF 3

DECK HALVES

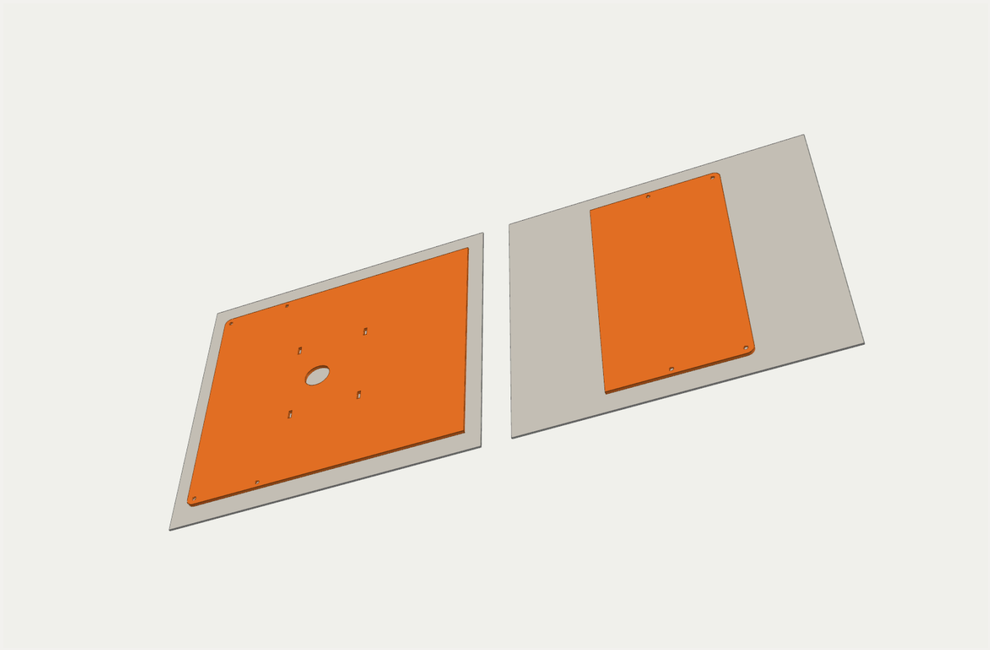


Bottom down, as exported. Both halves fit on one plate.

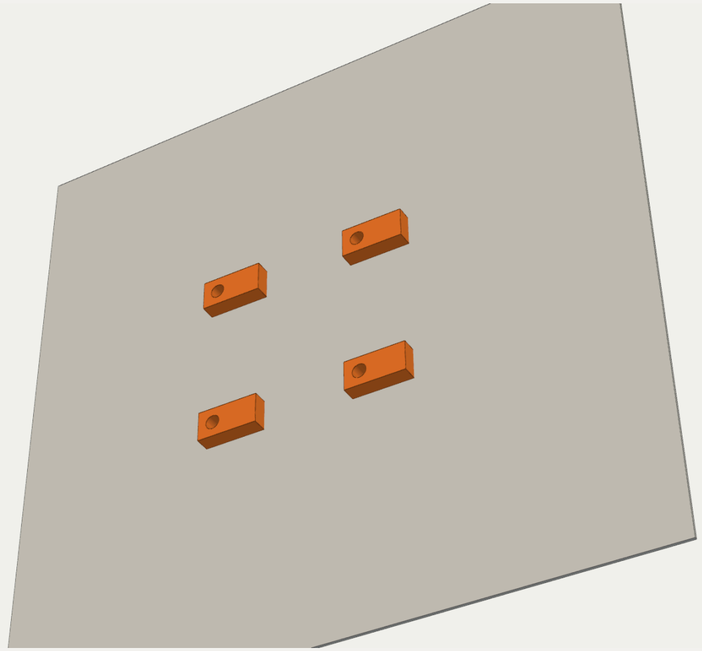
The bevelled back edges print as 45° chamfers. Those are what seat on the sleeve ledges and centre the deck.

> Keep elephant’s foot down on these edges. A fat first layer stops the deck seating fully.

BACK PLATE AND DOG-EAR ARMS



BACK PLATE L + R: flat, countersinks down. 20% gyroid.



DOG-EAR ARMS × 4: flat. Solid infill, 6 walls.

MACHINING · OP 1 OF 2

TRIM: BACK SIDE UP

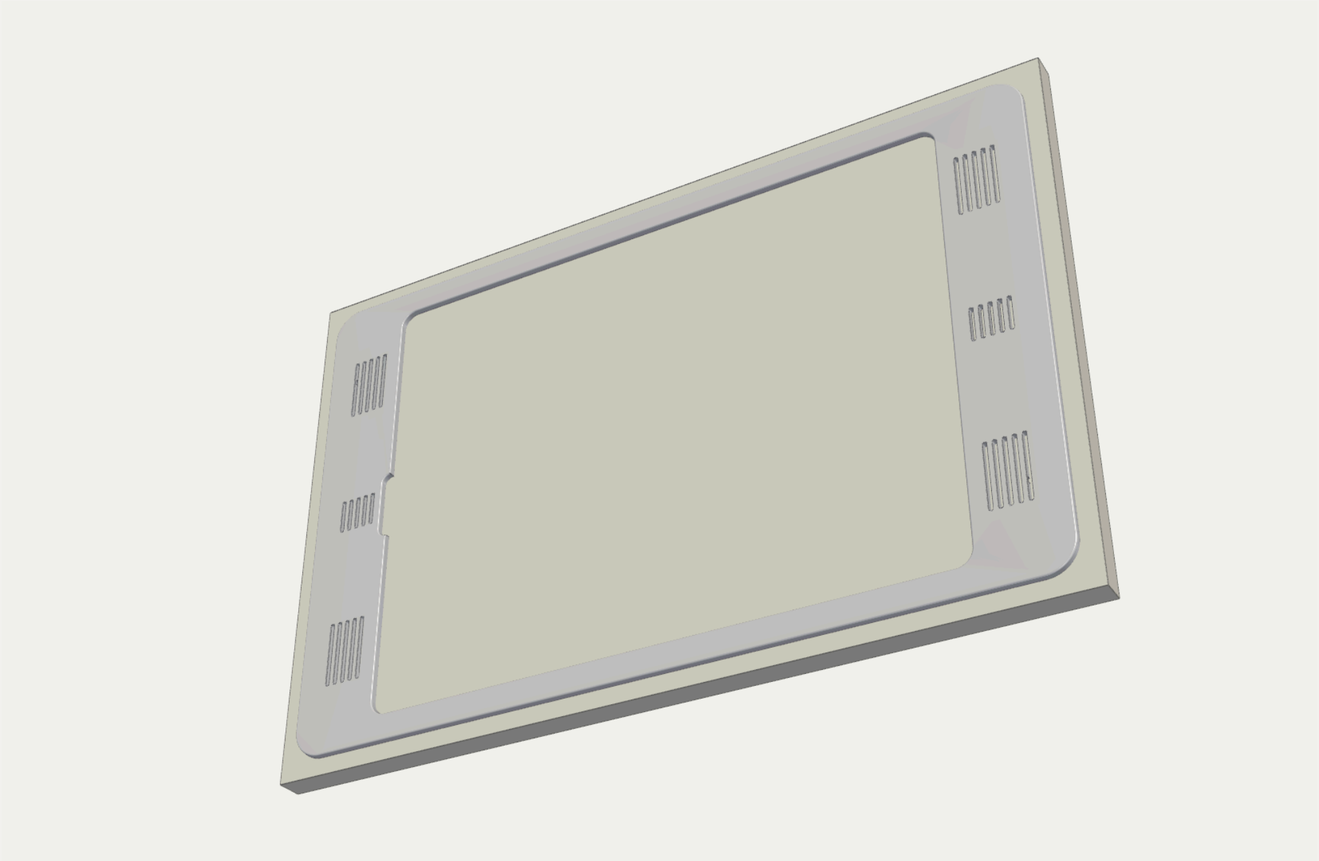


1. Hold 3 mm 6061 flat on a vacuum table or double-sided tape on a spoilboard.
2. Pocket 8 × Ø10.2 magnet seats, 1.6 mm deep, with a 6 mm end mill.
3. Drill 2 × Ø3 pin holes, 2 mm deep.
4. Cut the 0.55 mm plug relief at the USB-C end.

> **1.6 mm is the limit.** It leaves a 1.4 mm skin under the front face. Any deeper and the pocket can show through after blasting.

MACHINING · OP 2 OF 2

TRIM: FRONT SIDE UP



1. Flip it onto two Ø3 pins set in the spoilboard, using the pin holes from op 1.
2. Cut the window through: inner corners R7.5, Face ID stadium with R4.2 ends (6 mm end mill).
3. Cut the grille slots with a 2 mm end mill: climb, light passes.
4. Profile the outside with tabs, then a 0.8 mm radius on the front outer edge.

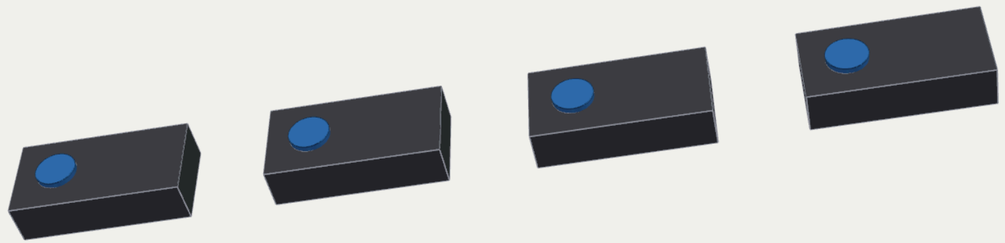
> **Finish:** fine glass bead blast, then clear or black anodise. Press in the 2 dowel pins after anodising.

01

BENCH ASSEMBLY

Ten steps on the bench. You finish with a complete in-wall unit (sleeve, deck, back plate, PoE adapter and arms) and a trim ready to fit.

ARM INSERTS



Press one **M4 insert** into each DOG-EAR ARM, from the top face.

Lay the arm flat on the bench so it can’t tilt. Push straight down and stop when flush.

> A crooked insert makes the arm swing out at an angle. Check each with a screw before it cools.

YOU NEED

4 × M4 heat-set inserts · iron with M4 tip

SLEEVE INSERTS



Press **8 M3 inserts** into the bosses on the back edge of the SLEEVE halves: 4 in each.

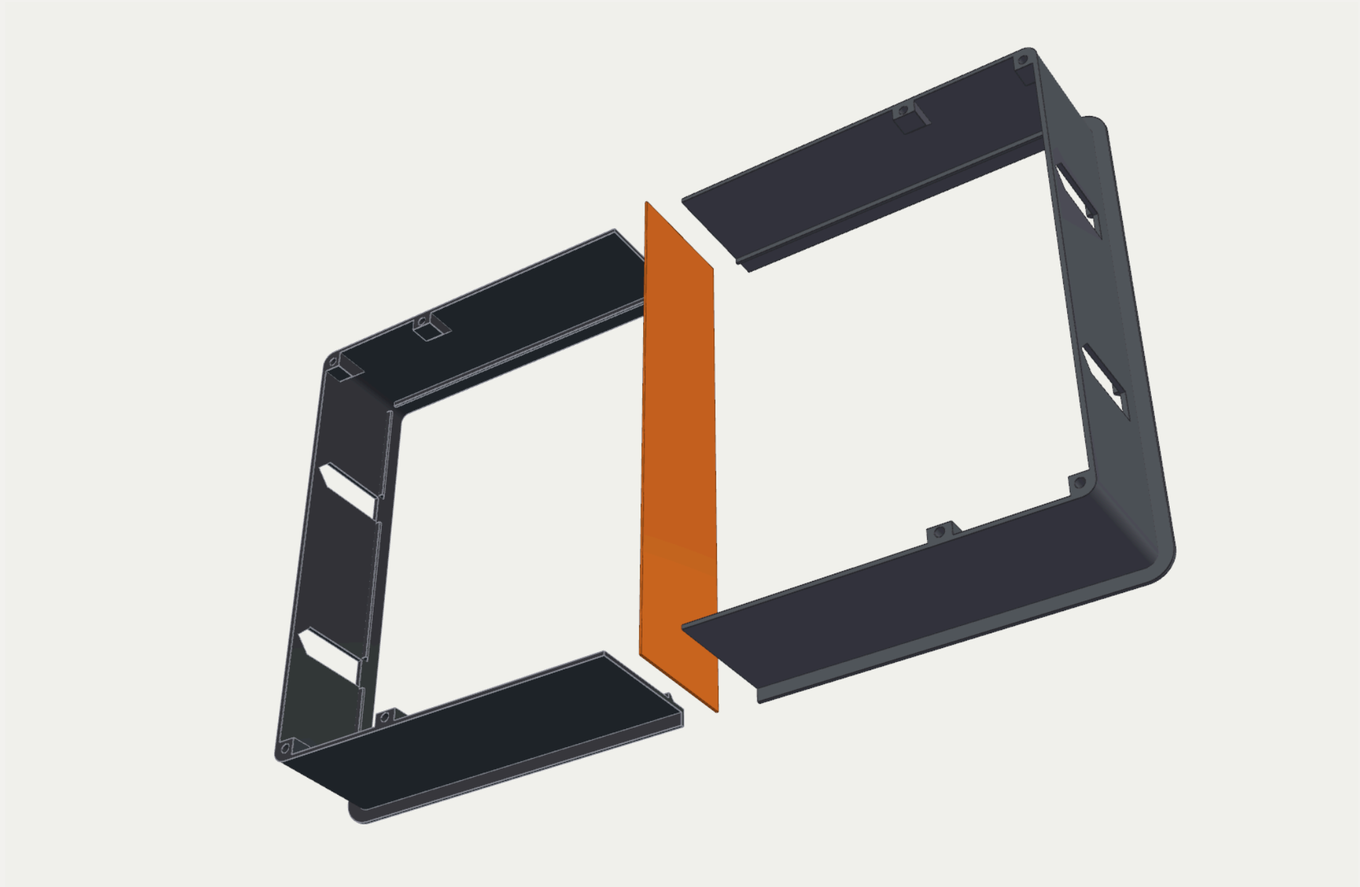
Do this before gluing. Each half sits flat on its flange, so you can press straight down.

> Stop each insert flush or 0.1 mm below. Proud inserts hold the back plate off its seat.

YOU NEED

8 × M3 heat-set inserts · iron with M3 tip

GLUE THE SLEEVE



Lay both SLEEVE halves **face down on a flat surface**. Epoxy the centre seam and push them together.

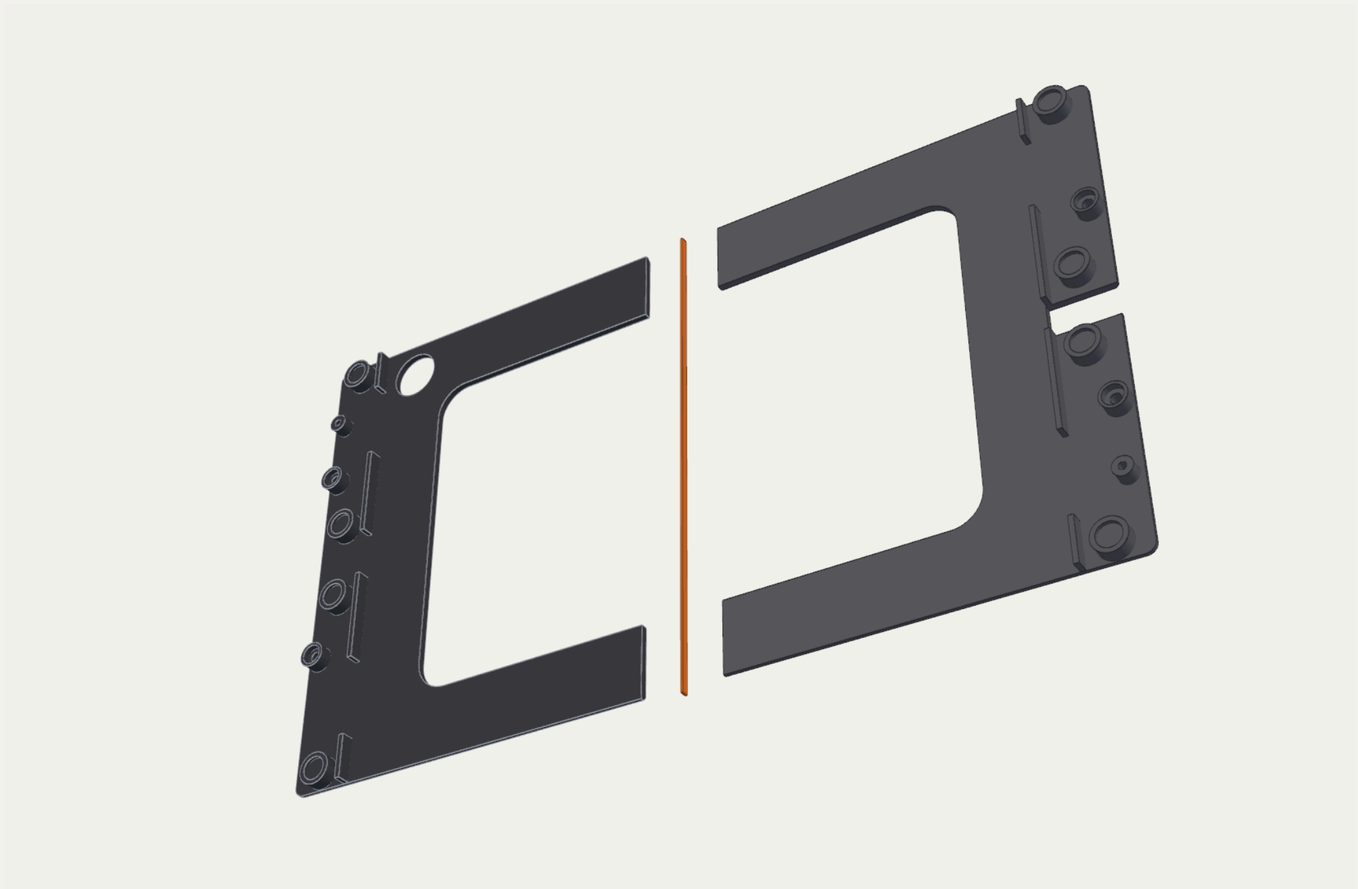
Keep the flanges pressed flat while it cures. Tape across the seam, inside and out, holds it closed.

> Wipe squeeze-out off the ledges. A blob there stops the deck seating.

YOU NEED

2-part epoxy · flat surface · masking tape

GLUE THE DECK



Epoxy the centre seam of the two DECK halves, **flat on the same surface**, and push them together.

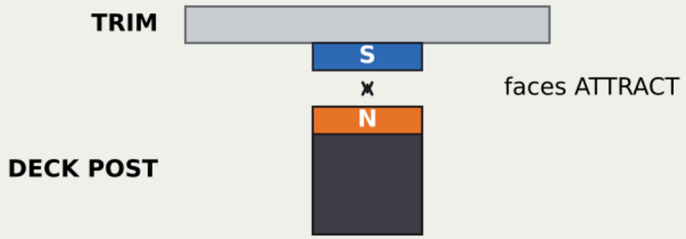
Once the sleeve and deck have both cured, dry-fit: the deck should drop onto the sleeve ledges and sit level with the flange.

> The iPad lies on this face. Sand any ridge at the seam flat, or it will press on the iPad’s back.

YOU NEED

2-part epoxy · flat surface · fine sandpaper

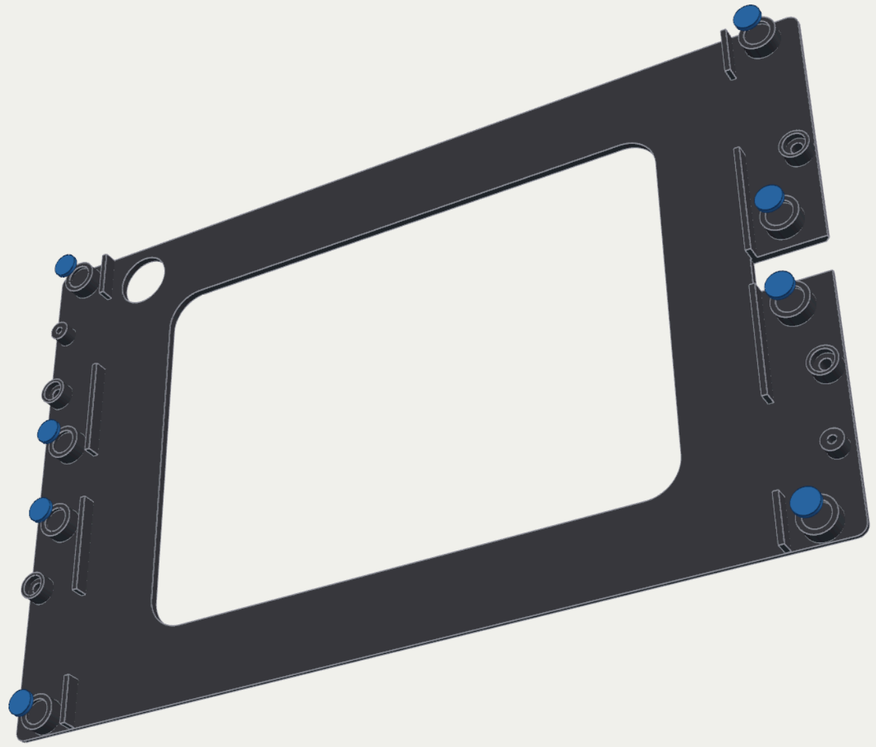
SORT THE MAGNET POLARITY



1. Stack all 16 magnets into one column.
2. Slide each off and dot its TOP face with a marker.
3. TRIM: dot face UP, into the pocket (toward the room).
4. DECK: dot face UP, exposed (toward the room).
   All 16 face the same way, so each pair attracts.
5. Test each pair before the glue sets: the trim must snap on.

> **Get this right before any glue comes out.** One reversed magnet and the trim pushes off on that corner. Every magnet goes in dot face toward the room.

DECK MAGNETS



Glue **8 magnets** into the MAGNET POSTS at the deck ends, **dot face up**.

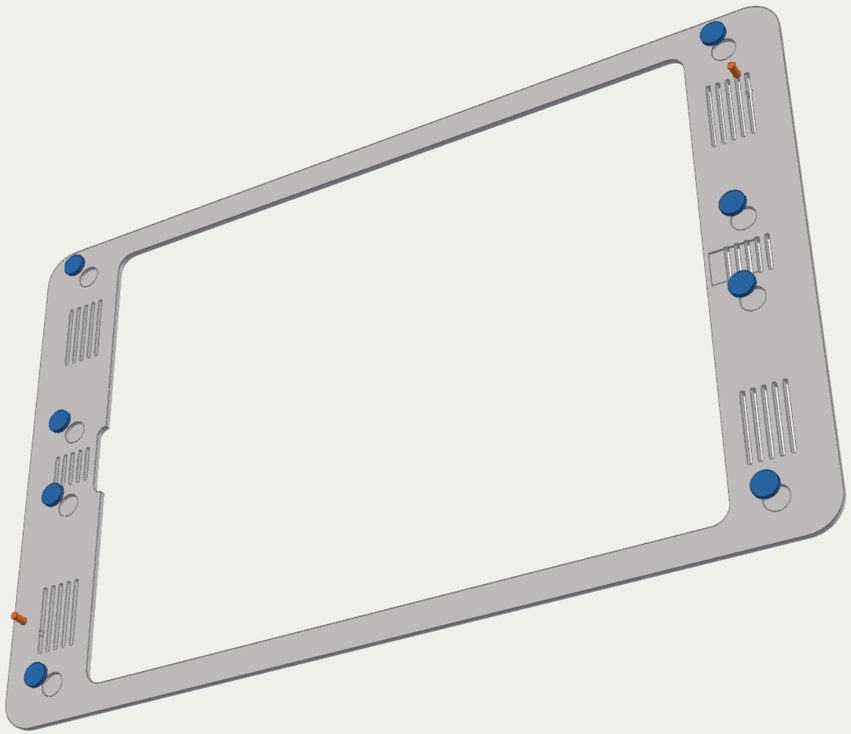
A small dab of epoxy in each seat. Press each magnet flush with a flat stick, then wipe off any excess.

> Fit them one at a time, well apart. Loose magnets jump to each other and to your tools.

YOU NEED

8 × N52 magnets 10 × 1.5 mm · epoxy

TRIM MAGNETS, PINS AND TAPE



Trim face down on a towel. Glue **8 magnets** into the pockets, **dot face into the pocket**. Press in the **2 dowel pins**.

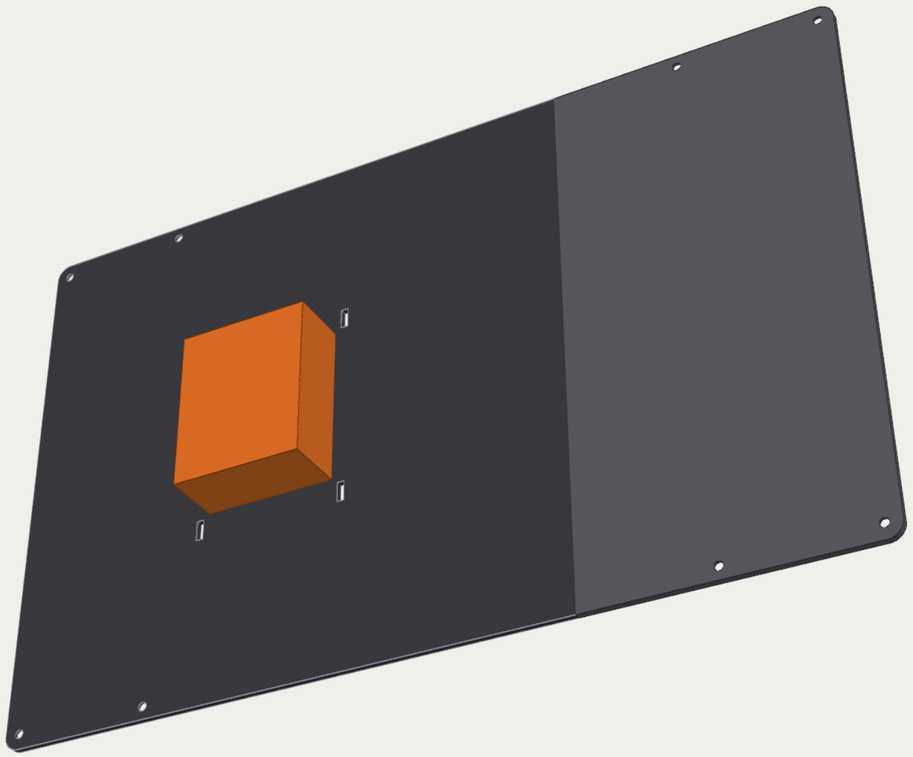
Then run PORON tape around the underside of the lip, where it will touch the glass.

> Before the glue sets, hold the trim over the deck. Every corner must pull in, not push away.

YOU NEED

8 magnets · 2 × Ø3 × 6 pins · epoxy · PORON tape

PoE ADAPTER ON THE BACK PLATE



Zip-tie the **PoE ADAPTER** to the inside face of the BACK PLATE through the slots. Plug in the USB-C cable.

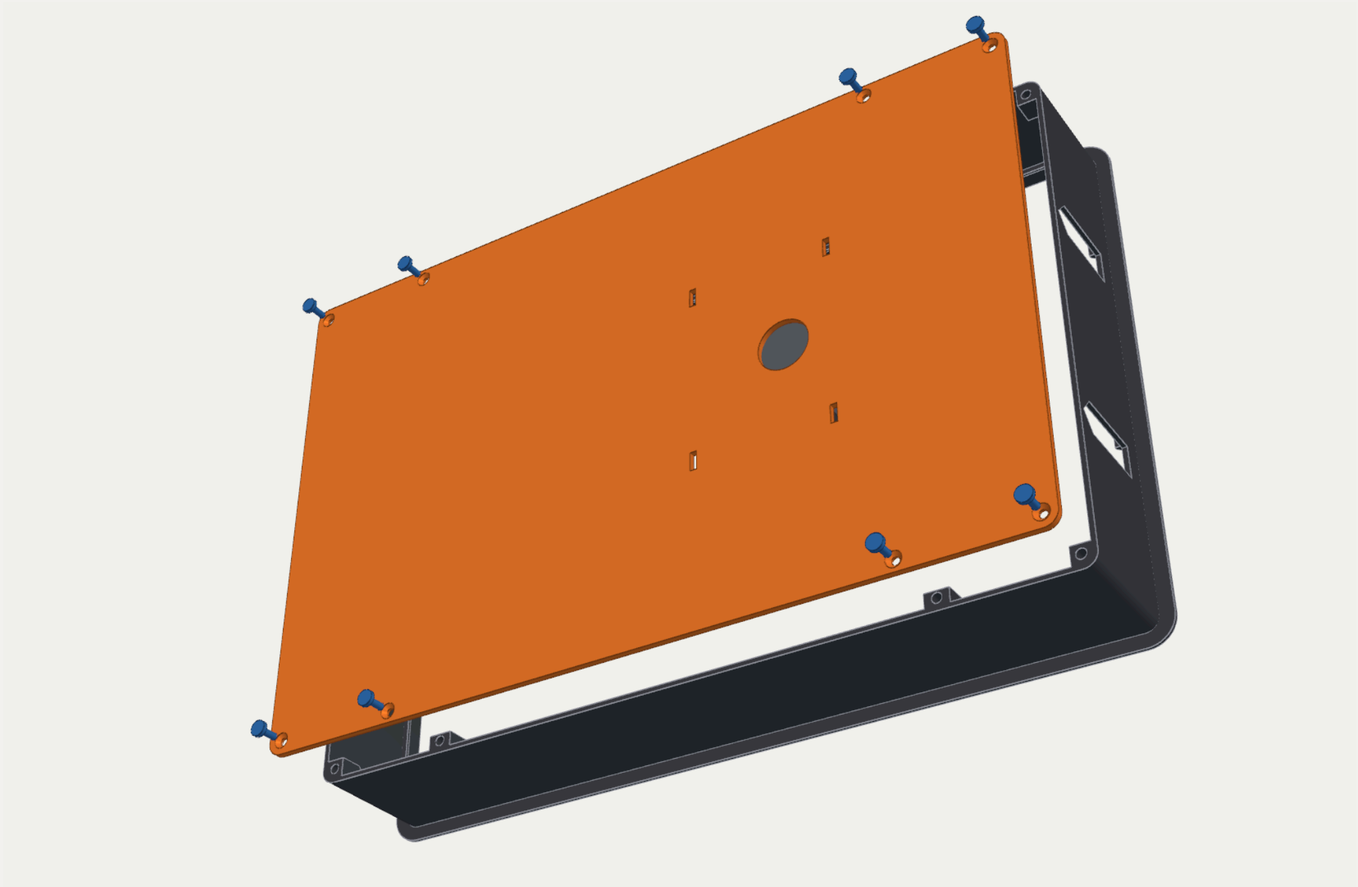
Point the adapter’s Ethernet socket at the Cat6 entry hole. Leave the USB-C cable loose for now.

> Snip the zip-tie tails flush so they don’t foul the deck.

YOU NEED

PoE adapter · 2 zip ties · USB-C cable

FIT THE BACK PLATE



Set both BACK PLATE pieces onto the sleeve’s back edge, adapter inside. Fit **8 × M3 × 8 countersunk** screws into the inserts.

Feed the USB-C plug up to the front of the box first, at the USB-C end.

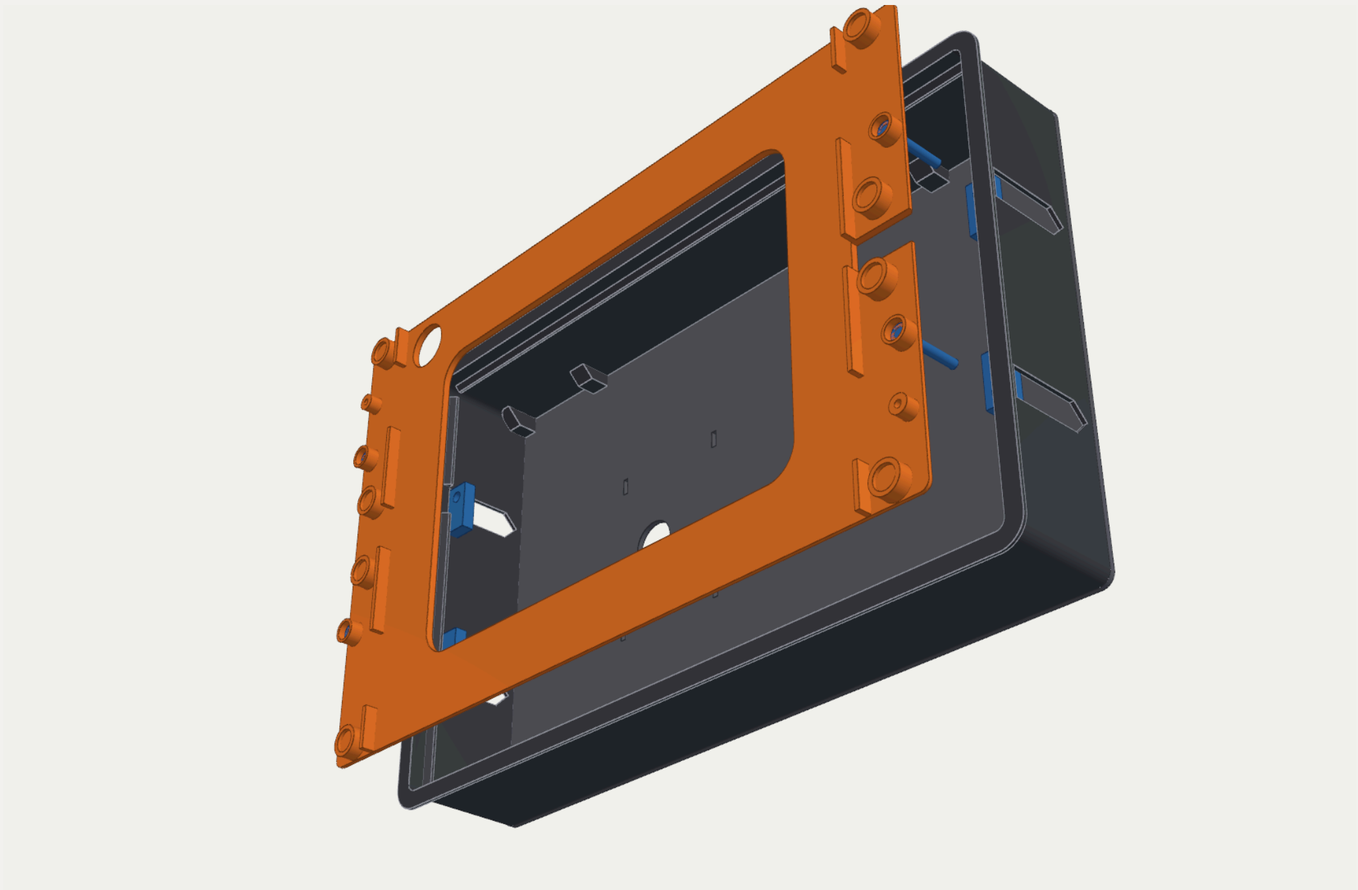
> The long piece crosses the sleeve seam. Snug the screws either side of the seam first.

YOU NEED

8 × M3 × 8 countersunk · 2 mm hex key

BENCH ASSEMBLY · STEP 10 OF 10

DECK AND ARMS IN



Bring the USB-C plug up through the gap in the ledge at the USB-C end, then drop the DECK onto its ledges.

Drive an **M4 × 30** down through each deck post and thread an ARM onto it inside its slot. Then back each screw out about 5 mm so the arms hang low.

> This is a trial fit. At the wall you lift the deck out once more to plug in the Cat6 (wall step 3).

YOU NEED

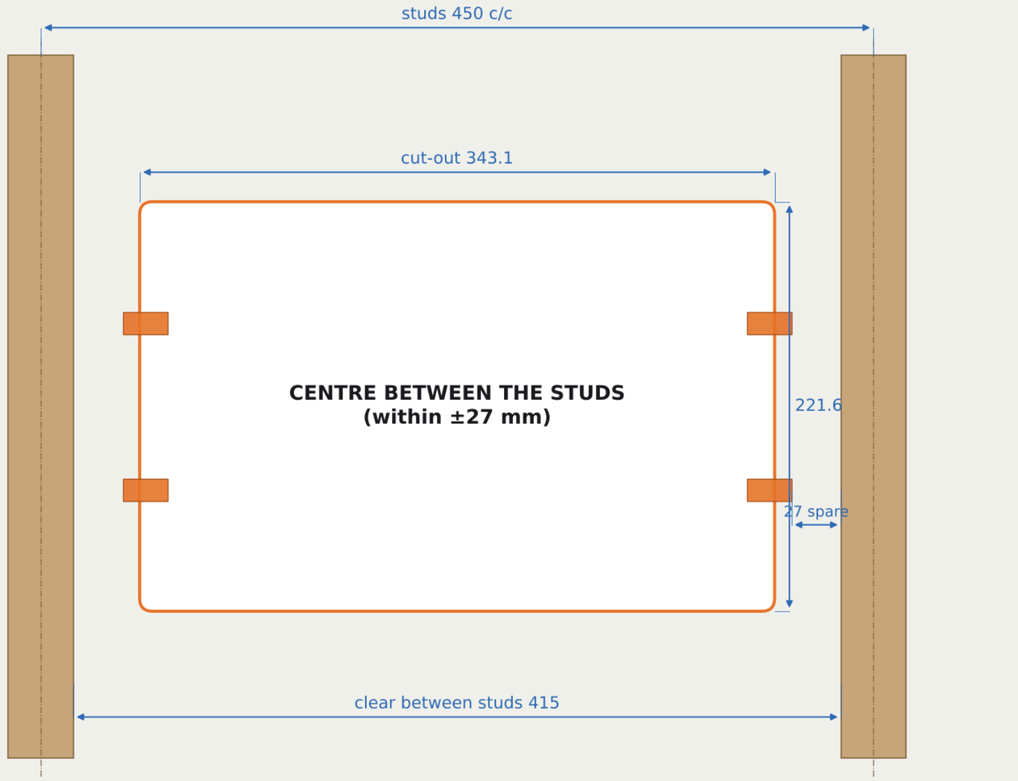
4 × M4 × 30 pan-head · 4 arms · screwdriver

02

WALL INSTALLATION

Seven steps at the wall. Measure twice: the cut-out is the only step you can’t undo.

FIND THE STUDS, PLACE THE TEMPLATE



Orange bars = dog-ear arms swung out (reach ±180.6 mm). Stud width assumed 35 mm (use 22 mm spare for 45 mm studs).

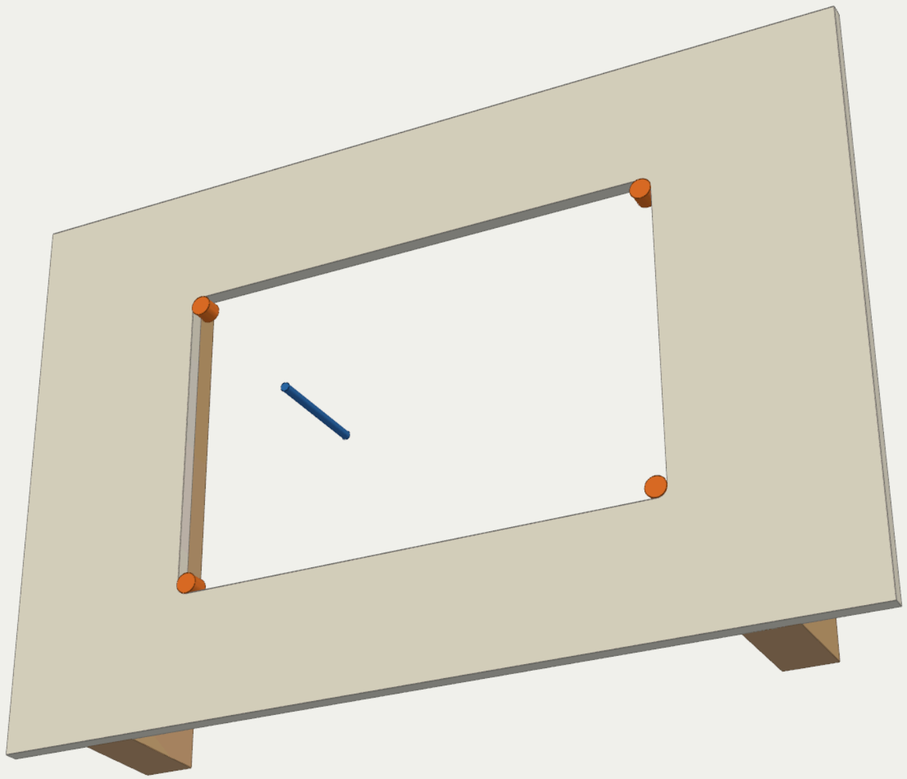
Mark both studs. Tape the **A3 TEMPLATE** up, level, **centred between them**.

The arms swing out sideways, so stay within 27 mm of the bay's centre (less with 45 mm studs). TOP ↑ on the template goes up.

Drill a small test hole in the middle first and look inside with a torch. Check for cables, pipes and noggings.

Print the template at 100% on A3. Its scale bar must measure exactly 100 mm.

CUT THE HOLE



Drill **Ø14 at the 4 corner crosses**. That gives the R7 corners.

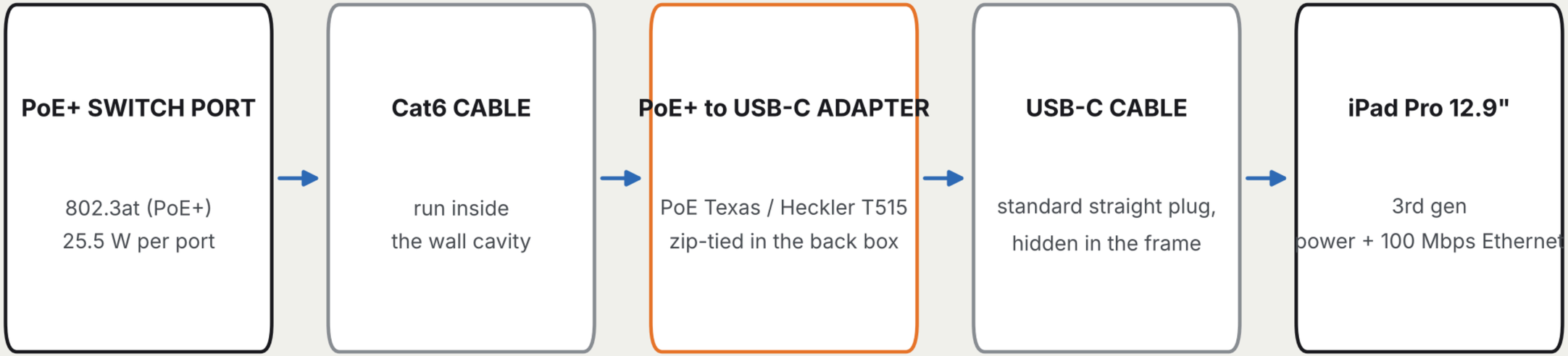
Saw the straight edges from hole to hole, on the solid line. Keep the saw square to the wall face.

> Err on the small side. The trim covers the cut edge by about 6 mm, so you can shave to fit but can’t add back.

YOU NEED

14 mm bit · plasterboard saw · dust mask

CONNECT THE CAT6

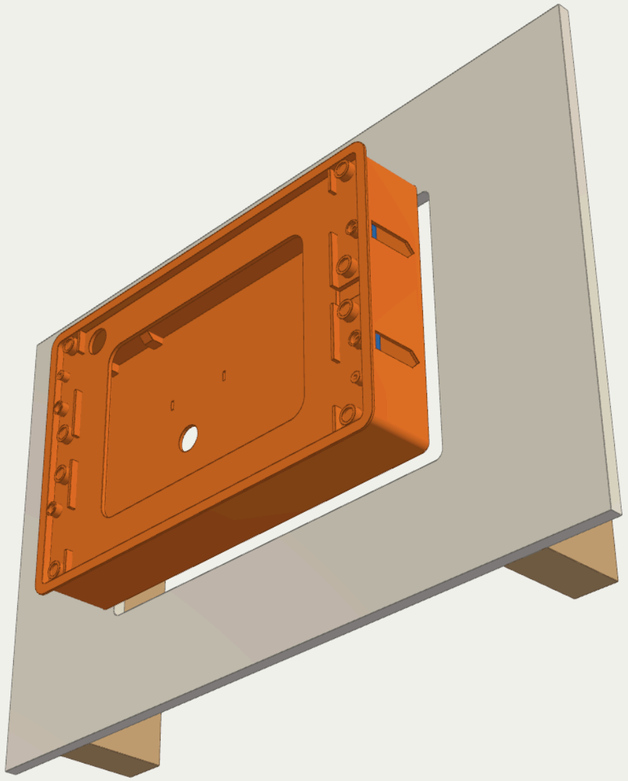


Fish the Cat6 out through the hole. Lift the DECK out: undo the 4 M4s and catch the ARMS as they come free.

Push the Cat6 in through the back plate’s entry hole and plug it into the PoE ADAPTER. Refit the deck, screws and arms as in bench step 10.

> Test now. Plug the iPad into the USB-C lead: it should charge and show Ethernet before the unit goes in the wall.

PUT THE UNIT IN THE WALL

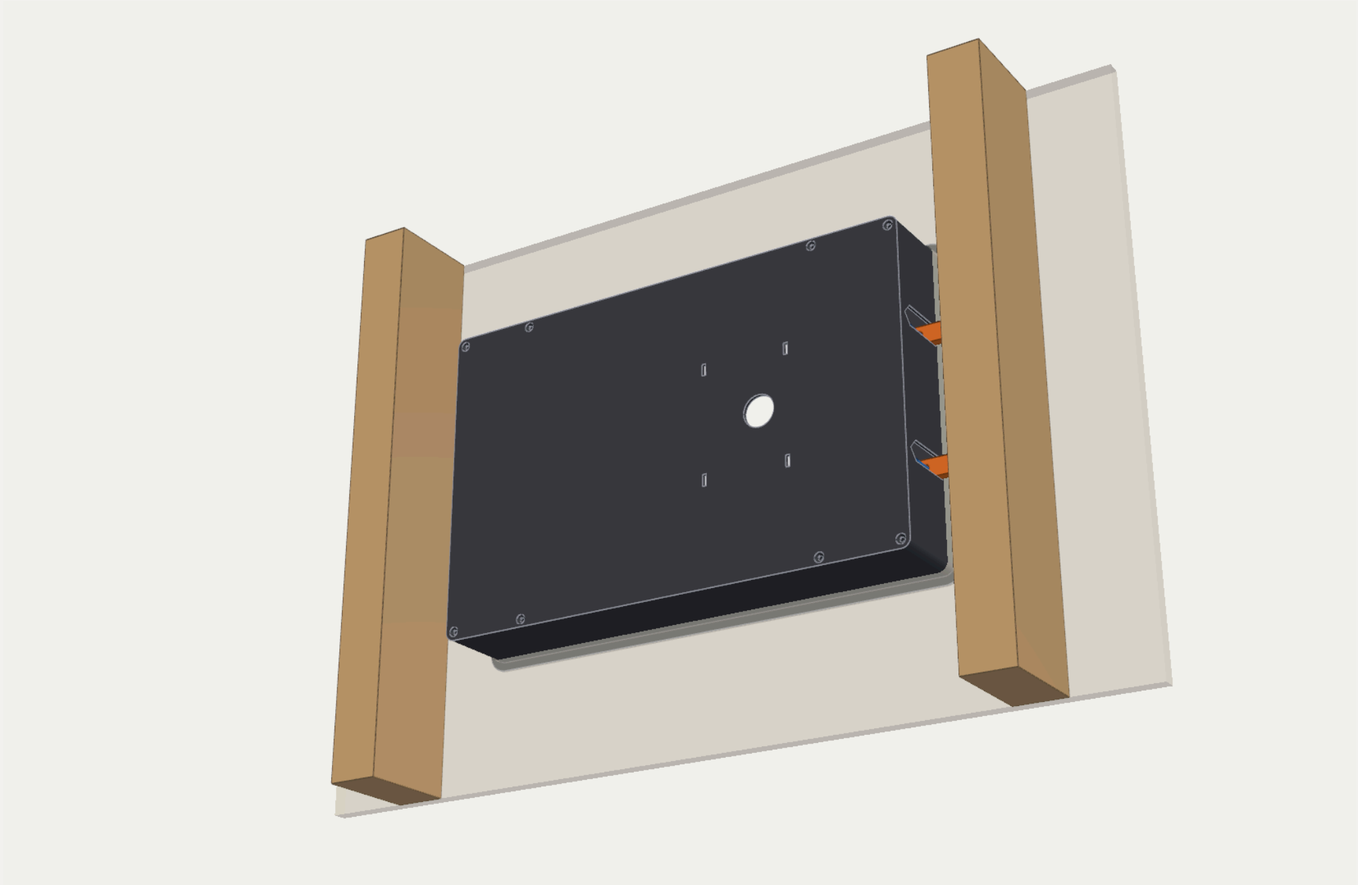


Turn all 4 ARMS to point into the box. Feed the Cat6 slack back into the wall, then push the unit straight in **until the flange sits on the wall**.

Keep the Cat6 entry on the side the template marks. The sleeve has 1 mm of clearance each side.

> Don’t force it. If it binds, shave the cut edge where it rubs.

CLAMP IT



Hold the flange flat on the wall and turn each **M4** clockwise. The ARM swings out through its slot, stops, then pulls up behind the board.

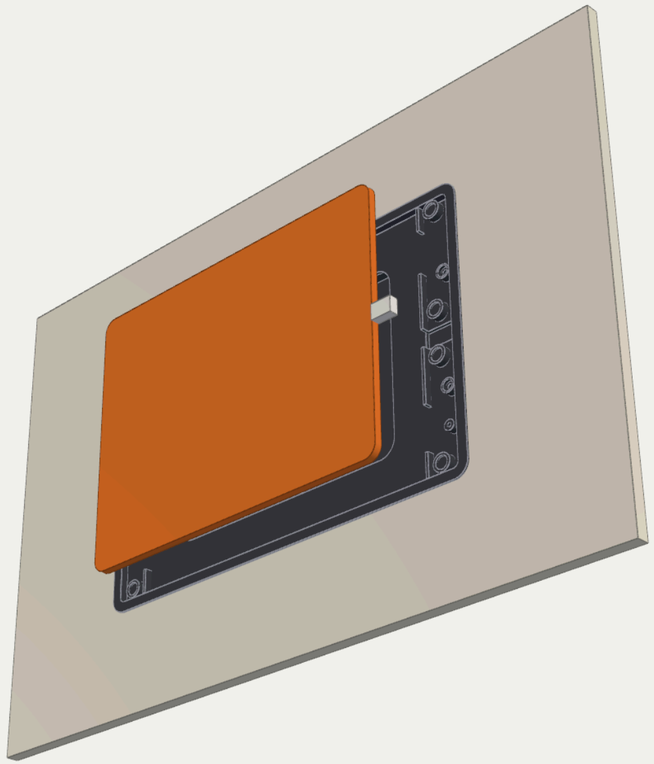
Tighten in diagonal pairs, a turn at a time, until the flange is snug all round.

> **Hand tight only.** Plasterboard crushes. Stop when the unit no longer moves.

YOU NEED

Hand screwdriver, not a drill

FIT THE iPAD

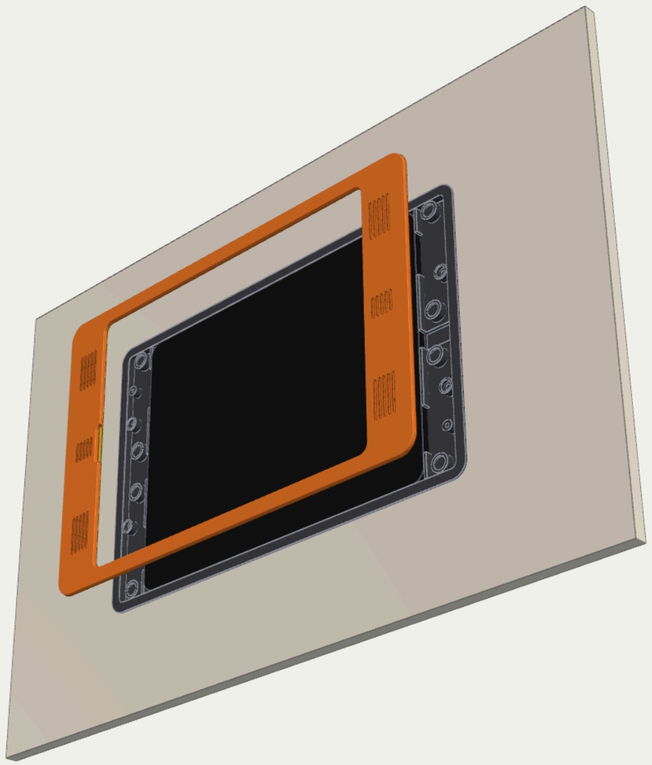


Plug the USB-C into the iPad. Lay it **screen out** on the DECK, plug end toward the cable gap.

The end ribs locate it side to side. Tuck the cable bend down into the box end.

> Clean the glass now. Once the trim is on, the edges are hard to reach.

FIT THE TRIM



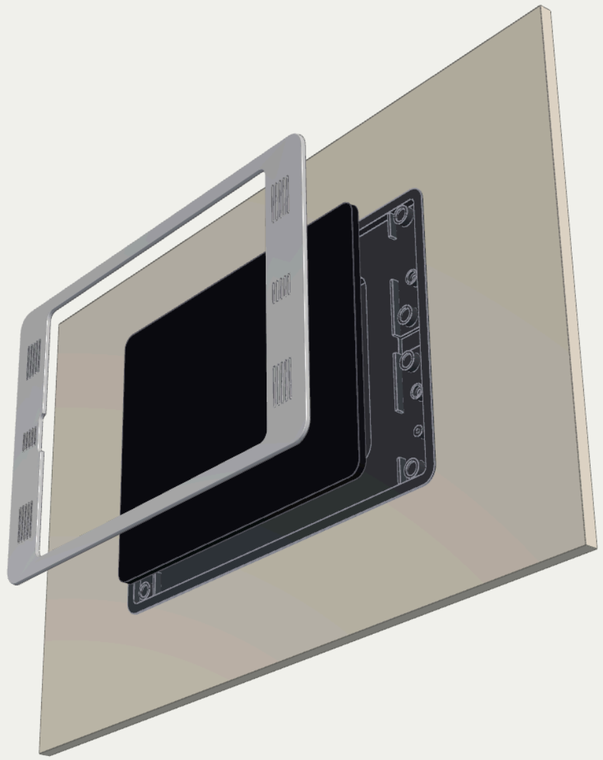
Line up the 2 PINS with their holes and offer the TRIM up. **The magnets pull it home.**

The pins also fit with the trim upside down, so check the Face ID cut-out sits over the camera first. The PORON lip rests on the glass edge.

> Keep fingers clear of the edges as it snaps on. Eight N52 pairs close hard.

SERVICE

REMOVING THE iPAD



Grip the TRIM by its wide ends and **pull straight out**. Lift the iPad off the deck and unplug it.

The deck stays put: the M4 screw heads hold it down.

> **Taking the unit out?** Back each M4 out a few turns until its arm swings in. Don’t remove the screws fully, or the arm drops into the wall.

SETUP

iPAD SETUP FOR A WALL PANEL

KEEP THE SCREEN ON

Settings › Display & Brightness › Auto-Lock › Never. Dim it at night with a Shortcut or Home Assistant automation.

LOCK TO ONE APP

Settings › Accessibility › Guided Access. Start it in Home or your dashboard app.

BUTTONS ARE COVERED

Turn on Settings › Accessibility › Touch › AssistiveTouch for lock, volume and restart. Siri works too.

MIND THE HEAT

The iPad vents into a closed box. Panel duty is fine; hours of video or games are not.

FINISHING BUILD

FINAL CHECKLIST

- [ ] Flange tight to the wall; unit doesn’t move
- [ ] Trim sits flat, no gap at any corner
- [ ] Face ID unlocks first time
- [ ] Speakers clear through the grilles
- [ ] iPad shows it is charging
- [ ] Ethernet appears in Settings
- [ ] No cracked plaster round the cut-out
- [ ] Guided Access and Auto-Lock set

> **Something off?** Every size lives in flush51.py. Measure the problem, change the parameter, re-run it, and reprint only the part that changed.

ENJOY YOUR FLUSH MOUNT

Feature positions on the iPad (buttons, cameras, speakers) are close approximations. Fit the printed parts and the iPad before you cut the wall or machine the trim.

Also available: Float Mount v4.1, the surface-mounted version.

